CS483 Profitable Project Proposal

Team Information

Team name: Profitable

Team members:

1) Khoi Le - kle539@uic.edu
2) Edgar Brizuela - ebriz2@uic.edu
3) alex stanson - astan27@uic.edu
4) Dillon Pathammavong - dillonfp@uic.edu

Problem and Goal

We want to build a model to predict stock prices for trading. Our goal is to have a model that exceed certain some average measures, such as Sharpe ratio, or return percentage

Potential Question to answer:

- Can a model built on mined data (price, news, filing features) from 2000-2020 beat SPY buy and hold on Sharpe ratio, after transaction costs, on unseen data from 2021 to 2023?

Data Plan

Necessary Data:

- Text: yahoo finance, company reports
- Numerical: stock price over time (Timeseries)

How:

- Data could be gathered either from public datasets or could be scraped from relevant websites. Examples:

Where: Time Estimate:

- FNSPID - S&P500 data prices and news [1999-2023] (GH, HF, Arxiv) ~ hours
- Kaggle dataset ~ hours
- sec.gov (filings, earnings releases, and footnotes) ~ 1-2 days

Data Mining Task

Methods:

- NLP features, TF-IDF on news and filings
- MinHash and LSH duplicate handling
- Spark MLlib to rank stocks weekly

Goals:

- Design a model that signals stock prices. Measure the performance on Sharpe ratio.
- Design a dashboard/terminal visualizing sentiment, signals and performance

Expected results:

- Public data will give small edge that is canceled out by trading costs
- Adding text sentiment to price data will be key to beating a price only model

Teamwork Agreement

Team Agreement

Development model

- Kanban (Agile) with one task board (To Do, Doing, Done)

Repo

- https://github.com/ebrizuela1/CS483-Project

Project Board

- https://github.com/users/ebrizuela1/projects/2

* course deadlines

Schedule

W8 Friday - Start (10/16)

W10 - Ensure data quality, collect data, test implementation and experimentation

W11 Sunday - Midterm check

W13 - Polish Repo, show methodology, ensure implementation quality

W14 - Slides, presentation prep and practice, final polishing

W15 Sunday - Final due